The Pattern Zoo

HackerRank C interview — recognition checklist. ~15 patterns cover the vast majority of screens. Recognition first; speed comes from use.

The 3-step reflex (under the clock):

1. **Read the bound on `n`** → that fixes the *allowed complexity* before you read the problem twice.
2. **Scan the cue column** → 2–3 patterns light up.
3. **The constraints kill the impossible ones** → usually one survives.

Pattern → Dead-giveaway cue

	PATTERN	DEAD-GIVEAWAY CUE
1	**Two-pointer**	sorted (or sortable) array, “find a pair/triple”, converge from both ends
2	**Sliding window**	“longest/shortest **contiguous** subarray/substring with property”, **no negatives**
3	**Prefix-sum array**	repeated “sum of range `[i,j]`” on a **fixed** array, 2D rectangle sums, “immutable” **(endpoints known)**
4	**Prefix sum + hashmap**	“contiguous sum = k”, endpoints unknown, especially **with negatives** **(window is dead)**
5	**Hashmap / freq array**	“count occurrences”, “seen before”, “distinct”, “anagram”
6	**Sorting + sweep**	“intervals/meetings overlap or merge”, “schedule”
7	**Binary search (array)**	sorted + “find/insert position of value”
8	**Binary search on answer**	“min/max value such that property holds”, “possible within budget K?” + monotonic
9	**Heap (priority queue)**	“kth largest/smallest”, “top k”, “running median”
10	**Monotonic stack**	“next greater/smaller element”, spans, histogram
11	**Stack**	“balanced”, “valid parentheses”, nesting, undo
12	**BFS**	“minimum steps to reach”, unweighted shortest path, levels
13	**DFS / union-find**	“connected components”, flood fill, “all paths”
14	**Dynamic programming**	“number of ways”, “min/max cost”, overlapping subproblems
15	**Math / bit trick**	huge `n`, divisibility/parity, “without extra space”

Constraint → complexity

Budget ~1e8 ops/sec. Work backwards from `n`.

N BOUND	ALLOWED TIME	LIKELY APPROACH
`≤12`	$O(n!)$ / $O(2^n \cdot n)$	brute force, backtracking
`≤25`	$O(2^n)$	subsets, bitmask DP, meet-in-middle
`≤500`	$O(n^3)$	Floyd-Warshall, interval DP
`≤5e3`	$O(n^2)$	2 nested loops, basic DP table
`≤1e6`	O(n log n)/O(n)	sort, two-pointer, window, hashing, heap
`≤1e8`	O(n)	single pass, prefix sums, counting
`≤1e18`	O(log n)/O(1)	binary search, math/closed form

When stuck — 3 questions

ASK	IF YES →
Can I **sort** it?	two-pointer, sweep, binary search
Can a **hashmap** make lookup O(1)?	freq count, “seen before”, prefix-sum
Can I keep a **running** quantity in one pass?	window, prefix sum, counting

C gotchas (the ❌ on submit)

- `long long` for sums > ~2e9; print `%lld`.
- `mid = lo + (hi-lo)/2` — never `(lo+hi)/2`.
- Comparator `(a>b)-(a<b)`, not `a-b` (overflow).
- Off-by-one on 1-indexed input.
- Consume newline before `fgets` after `scanf("%d")`.